GuidePoint — User Handbook

GuidePoint — User & Developer Handbook

Indoor QR navigation with voice-guided routing

Document version	1.1 (May 2026) · pairs with repo release **1.3.4**
Repository (web)	https://github.com/nandanpabolu/GuidePoint_NAN
Clone (HTTPS)	`https://github.com/nandanpabolu/GuidePoint_NAN.git`
Maintainer affiliation	The University of Texas at Dallas · UTDesign EPICS (Computer Science)

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1. What GuidePoint Does

GuidePoint helps people navigate indoors when GPS is unreliable:

1. A **QR code** provides a building **floor graph** as structured **JSON** (either embedded directly in the QR or loaded from an **HTTPS** URL printed in the QR).
2. After the map loads, the traveler **speaks a destination** (e.g. “Seminar Hall”). The app matches speech to waypoint **names** in the graph.
3. **A* pathfinding** computes a walking route from a **starting node** to the requested place.

4. The **navigation** screen gives **spoken (TTS) turn-by-turn** hints, optionally uses **step counting** between nodes, and can use optional **TensorFlow Lite** vision models where configured.

There is **no built-in backend server**: maps are ordinary JSON documents. Facilities partners may host maps on **any HTTPS file host** when using URL-based QR payloads.

Full technical overview: root **`README.md`** in the repository and **`docs/ARCHITECTURE_SENSOR_FUSION.md`** (engineering deep dive).

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2. Who This Guide Is For

Audience	Typical use of this handbook
End users / testers	Install a build someone gives you, grant permissions, scan QR, speak destination → follow cues. Sections 5–9.
Developers & EPICS collaborators	Clone the repo, install Flutter toolchain, run from source, run tests. Sections 3–11.
Facilities partners	Prepare JSON graphs and QR rollouts (**`docs/MAP_DATA_SCHEMA.md`**). Sections 10–11.

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3. Repository Layout (Important Paths)

After you clone **`https://github.com/nandanpabolu/GuidePoint_NAN.git`**, useful locations are:

Path	Contents
`flutter_app/`	Flutter project — primary app (**`flutter run`** from here).
`tools/scan_atl_qr.html`	Browser page with an **ATL sample-map QR** for quick testing. Open this file locally in Chrome/Safari/Edge after cloning.
`data/maps/ATL_JSON.json`	Example building map aligned with **`docs/MAP_DATA_SCHEMA.md`**.
`docs/`	Setup (**`SETUP_FLUTTER_AND_RUN.md`**), emulator (**`ANDROID_EMULATOR_SETUP.md`**), demos (**`DEMO_AND_TESTING.md`**), map schema (**`MAP_DATA_SCHEMA.md`**).

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4. Prerequisites (Development / QA)

Minimum for **running from source**:

- **Flutter SDK** 3.2+ (`https://docs.flutter.dev/get-started/install`)
- **Dart** (bundled with Flutter)
- For **Android**: Android SDK / Android Studio, USB debugging on device or emulator
- For **optional iOS** builds on macOS: Xcode, Apple tooling
- A **physical phone** recommended for microphone, realistic camera QR scanning, and **pedometer** testing (emulators often approximate these poorly)

Always verify toolchain health:

```
flutter doctor -v
```

Accept Android SDK licenses when prompted:

```
flutter doctor --android-licenses
```

First-time setup walkthrough without prior Flutter experience: **`docs/SETUP_FLUTTER_AND_RUN.md`** in the cloned repository.

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5. Get the Source and Run (Developers)

```
git clone https://github.com/nandanpabolu/GuidePoint_NAN.git
cd GuidePoint_NAN
cd flutter_app
flutter pub get
flutter analyze
flutter test
flutter run
```

When **`flutter run`** lists devices, pick your phone or emulator. On first launch, **accept Camera** and **Microphone** permissions when prompted.

macOS desktop (fast demo, fewer sensor limitations for layout testing):

```
cd flutter_app
flutter run -d macos
```

Deep demo script: **`docs/DEMO_AND_TESTING.md`** (same repo, `docs/` folder).

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6. First Launch — End-User Screens

6.1 Terms & conditions

First launch shows **Terms & Conditions**. Acceptance is stored (**SharedPreferences**) so returning users skip this after tapping **Accept**.

6.2 QR scanner

The app opens **camera preview** with **QR decoding**. You must scan a QR whose payload is either:

- Valid **inline JSON** for the navigation graph (compact maps only — very large JSON may exceed practical QR density), OR
- A **literal HTTPS URL string** pointing to JSON (**GET** fetch)

After a successful read, the app parses nodes/edges (**`docs/MAP_DATA_SCHEMA.md`**) into the pathfinding engine.

6.3 “Say your destination”

Once the graph is loaded, the app listens via **speech-to-text**. Speak one of the **node display names** in the JSON (substring matching is supported). Examples that match the bundled ATL demo: **“Seminar Hall”**, **“Idea Labs”**, **“Hallway Junction”**.

6.4 Navigation & instructions

The **Navigation** screen:

- Gives **spoken** distance/next-step guidance (**TTS**) on an interval cadence
- Tracks **Steps** via **`pedometer`** on capable hardware
- Optionally uses **camera** + bundled **TensorFlow Lite** assets for landmark confirmation — behavior depends on which **`.tflite`** files are present under **`flutter_app/assets/models/`**

6.5 Instruction replay

Use the **list icon** on the navigation app bar (**Stored data** route) to reread textual instructions.

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7. Obtaining an ATL Demo QR Quickly

1. Clone or download the repo.
2. On your laptop, open **`tools/scan_atl_qr.html`** from the repository root in any modern browser — a large QR renders on screen.
3. With **GuidePoint** running on a phone (**QR scanner** screen), scan that QR from the laptop display (brightness up; hold steady).

If microphone permission fails, revisit **Android Settings → Apps → GuidePoint → Permissions**.

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8. Alternate Demos Without a Printed QR or Camera Issue

During **development** builds (**debug/profile**):

1. From the scanner screen, tap **"Load sample map"** → loads bundled sample JSON (ATL-style).
2. On the subsequent screen, tap **"Preview navigation UI"** → opens navigation UI with a **fixed sample route**.

These shortcuts are omitted in **release** consumer builds (**`flutter build`** release). Prefer real QR payloads for stakeholder demos that mimic production behavior.

Guidance without a physical Android device is in **`docs/ANDROID_EMULATOR_SETUP.md`**; use emulator camera → laptop webcam routing if scanning from the emulator.

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9. Designing Your Own Test QR Payloads

9.1 Inline JSON QR

Minimal structure (floors contain **nodes** and **edges**; each node includes **`id`**, **`name`**, **`position`**):

See **`docs/DEMO_AND_TESTING.md`** for a copy-paste small graph.

9.2 Optional start override

Include root key **`start_node_id`** when the sticker is glued at a waypoint other than the default graph entry:

```
{
  "building": { "name": "MyBuilding", "floors": [ ... ] },
  "start_node_id": "junction_1"
}
```

9.3 URL-based QR

If the QR string is **`https://example.com/maps/site.json`** the app **`GET`s** JSON and parses it exactly like inline content. Use **HTTPS**.

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10. Verification Checklist (QA Before Demos)

#	Step	Expected result
1	`flutter analyze` inside **`flutter_app/`**	Reports **No issues**.
2	`flutter test`	All tests green.
3	Install debug build (`flutter run`)	App launches Home → QR flow.
4	Scan **`scan_atl_qr.html`**	Map loads → STT activates.
5	Say **"Seminar Hall"**	Route exists → navigation cues play.
6	Deny mic deliberately	Recover by enabling permission + retry speech.

Extended matrix: **`docs/DEMO_AND_TESTING.md`** §5–6.

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11. Facilities Partner Notes (Maps & Hosting)

Author JSON using **`docs/MAP_DATA_SCHEMA.md`** as canonical structure. Maintain:

- Stable **`id`** values for nodes (programmatic references) vs human **`name`** strings (speech targets).
- **`edges`** with realistic **`distance`** units (consistent with cues).
- Printed QR placement near graph entry (**`start_node_id`** when appropriate).

Hosting only requires static file distribution or CDN — no database prerequisite.

Sample graph file: **`data/maps/ATL_JSON.json`**.

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12. Troubleshooting Summary

Symptom	Things to verify
QR unreadable	Brightness/contrast on screen QR; shorten JSON vs URL-hosted map
`flutter` not found	Install Flutter SDK; reopen terminal; **`flutter doctor`**
No devices listed	USB debugging enabled; emulator started; **`adb devices`**
Silent STT path	Microphone granted; noisy room; articulate destination phrase
"Could not find place named…"	Spoken string must loosely match waypoint **`name`** field
Steps frozen at zero	Real device sensors; emulator limitations
`scene_classifier`/YOLO not active	Expect limited confirmation without matching **`.tflite`** in **`assets/models/`** + rebuilt app

Detailed table: **`docs/DEMO_AND_TESTING.md`** §6.

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13. From Development Build Toward Stores (Brief)

Publication is **outside** this handbook's procedural scope but the release path uses standard Flutter signing:

Platform	Outline
Android Play	Create your release **keystore**. Copy **`flutter_app/android/key.properties.example`** → **`flutter_app/android/key.properties`** and fill in real passwords and paths (**never commit** those secrets). Then run **`flutter build appbundle`** — see **`https://docs.flutter.dev/deployment/android`**.
iOS	Enroll in the **Apple Developer Program**, set up **Xcode** signing profiles, then run **`flutter build ipa`** — see **`https://docs.flutter.dev/deployment/ios`**.

Official references:

- Android: **`https://docs.flutter.dev/deployment/android`**
- iOS: **`https://docs.flutter.dev/deployment/ios`**

High-level bullets also appear in repo root **`README.md`**.

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14. Project Team — UT Dallas (EPICS Computer Science)

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Partner institutions and extended collaborators coordinate through EPICS stakeholder channels alongside **BVRIT** engineering teams when joint deployments arise.

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15. License

Software is distributed under the **MIT License** — **`LICENSE`** at repository root:
https://github.com/nandanpabolu/GuidePoint_NAN/blob/main/LICENSE

Appendix A — Canonical Links Inside the Repository

Topic	Relative path after clone
Product & engineering README	`README.md`
Flutter subproject notes	`flutter_app/README.md`
Demo & feature tests	`docs/DEMO_AND_TESTING.md`
First Flutter install	`docs/SETUP_FLUTTER_AND_RUN.md`
Android emulator setup	`docs/ANDROID_EMULATOR_SETUP.md`

Topic	Relative path after clone
Map schema	`docs/MAP_DATA_SCHEMA.md`

Repository home: https://github.com/nandanpabolu/GuidePoint_NAN

End of handbook.